Section 4 - Practice Problems

Matthew Farah, ⟨farahm11@mcmaster.ca⟩, 400468251

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Chapter 1: Limits of Functions

Exercise 4.1.4

Let $f \coloneqq \mathbb{R} \to \mathbb{R}$ and let $c \in \mathbb{R}$. Show that $\lim_{x\to c} f(x) = L$ if and only if $\lim_{x\to 0} f(x+c) = L$.

Solution:

(I) Prove the forward direction.

By definition of functional limits, we know that if $\lim_{x\to c} f(x) = L$, then for arbitrary $\epsilon > 0$, there exists a $\delta > 0$ such that $0 < |x - c| < \delta \implies |f(x) - L| < \epsilon$.

$$\begin{aligned} 0 < |x-0| < \delta \implies \\ |f(x+c) - L| &= |f(x+c) - f(x) + f(x) - L| \\ &\leq |f(x+c) - f(x)| + |f(x) - L| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon \end{aligned}$$

(II) Prove the reverse direction.

Exercise 4.1.5

Let $I \coloneqq (0, a)$ where $a > 0$, and let $g(x) \coloneqq x^2$ for $x \in I$. For any points $x, c \in I$, show that $|g(x) - c^2| \leq 2a|x - c|$. Use this inequality to prove that $\lim_{x\to c} x^2 = c^2$ for any $c \in I$.

Solution:

(I) Prove that $|g(x) - c^2| \leq 2a|x - c|$ for all $x,\ c \in I$.

$$\begin{aligned} |g(x) - c^2| &= |x^2 - c^2| \\ &= |\,(x+c)\,(x-c)\,| \\ &\leq |\,(a+a)\,(x-c)\,| \\ &= |2a\,(x-c)\,| \\ &= 2a|x-c| \end{aligned}$$

(II) Prove that $\lim_{x\to c} x^2 = c^2$ for all $c \in I$.

Let $\epsilon > 0$ be arbitrary. Selecting $\delta = \frac{\epsilon}{2a}$ we can see that, in conjunction with the inequality derived earlier, that for all $x \in I$ satisfying $0 < |x - c| < \delta$:

$$\begin{aligned} |g(x) - c^2| &\leq 2a|x-c| \\ &< 2a \cdot \frac{\epsilon}{2a} \\ &= \epsilon \end{aligned}$$

Thus, by definition of functional limits, we know that $\lim_{x\to c} g(x) = \lim_{x\to c} x^2 = c^2$ for arbitrary $c \in I$.

Exercise 4.1.8
Show that $\lim_{x\to c} \sqrt{x} = \sqrt{c}$ for any $c > 0$.

Solution:

Let $\epsilon > 0$ be arbitrary. Selecting $\delta_0 = \sqrt{c}$, we can see that:

$$0 < |x - c| < \delta_0 \implies |\sqrt{x} + \sqrt{c}| > c$$

Since $c - \sqrt{c} < x < c + \sqrt{c}$ must hold for $|x - c| < \delta_0$ to be satisfied. Now consider $\delta_1 = c\epsilon$ and setting $\delta = \min \delta_0,\, \delta_1$. We therefore have for all x satisfying $0 < |x - c| < \delta$:

$$\begin{aligned}
|\sqrt{x}-\sqrt{c}| &= |\sqrt{x}-\sqrt{c}| \cdot \frac{\sqrt{x}+\sqrt{c}}{\sqrt{x}+\sqrt{c}} \\
&= \frac{|x-c|}{\sqrt{x}+\sqrt{c}} \\
&< \frac{\delta}{c} \\
&\leq \frac{\delta_0}{c} \\
&= \frac{c\epsilon}{c} \\
&= \epsilon
\end{aligned}$$

Therefore, by definition of functional limits, $\lim_{x\to c}\sqrt{x} = \sqrt{c}$.

Exercise 4.1.12

Show that the following limits do *not* exist.

(a) $\lim_{x\to 0} \frac{1}{x^2}$ $(x > 0)$.

Solution:

To prove that $\lim_{x\to 0}$ does not exist, we must prove that for all $L \in \mathbb{R}$, there exists an $\epsilon > 0$ such that for all $\delta > 0$ there always exists some x_0 satisfying $|x - 0| < delta$ such that $|\frac{1}{x_2} - L| \geq \epsilon$. Thus,

By the Archemedian principle, we know that there exists an $N \in \mathbb{N}$ such that $N > \sqrt{\delta}$. Thus, $N^2 > \delta$ and thus $\frac{1}{N^2} < \frac{1}{\delta}$

$$\begin{aligned}
0 < |x| < \delta \implies |\frac{1}{x^2} - L| &= |\frac{1 - Lx^2}{x^2}| \\
&=
\end{aligned}$$

(b) $\lim_{x\to 0} \frac{1}{\sqrt{x}}$ $(x > 0)$.

Solution:

(c) $\lim_{x\to 0}(x + \text{sign}(x))$. Note that $\text{sign}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$.

Solution:

(d) $\lim_{x\to 0} \sin\left(\frac{1}{x^2}\right)$.

Solution:

Exercise 4.1.14

Let $c \in \mathbb{R}$ and let $f : \mathbb{R} \to \mathbb{R}$ be such that $\lim_{x\to c}(f(x))^2 = L$.

(a) Show that if $L = 0$, then $\lim_{x\to c} f(x) = 0$.

Solution:

If $L = 0$, then for arbitrary $\epsilon > 0$, there exists a $\delta > 0$ such that all x satisfying $0 < |x-c| < \delta$ also satisfies $|\left(f(x)\right)^2 - L| < \epsilon^2$. However, note that if $L = 0$, then:

$$\begin{aligned} |\left(f(x)\right)^2 - L| &< \epsilon^2 \\ |\left(f(x)\right)^2 - 0| &< \epsilon^2 \\ |\left(f(x)\right)^2| &< \epsilon^2 \\ |f(x)|^2 &< \epsilon^2 \\ |f(x)| &< \sqrt{\epsilon^2} \\ |f(x)| &< \epsilon \\ |f(x) - 0| &< \epsilon \end{aligned}$$

Thus, since for all $\epsilon > 0$, there exists a $\delta > 0$ such that all x satisfying $0 < |x - c| < \delta$ also satisfies $|f(x) - 0| < \epsilon$, then by definition of functional limits, $\lim_{x\to c} f(x) = L$.

(b) Show by example that if $L \neq 0$, then f may not have a limit at c.

Solution:

If $\left(f(x)\right)^2 = x$, we can trivially see that $\lim_{x\to 1}\left(f(x)\right)^2 = \lim_{x\to 1} x = 1$ and $\lim_{x\to 1} f(x) = \lim_{x\to 1}\sqrt{x} = 1$. However, if $\left(f(x)\right)^2 = \frac{1}{x-1}$ then $\lim_{x\to 0}\left(f(x)\right)^2 = \lim_{x\to 0}\frac{1}{x-1} = -1$, however since $f(x) = \frac{1}{\sqrt{1}x-1}$ has an interval around $x = -1$ for which all x values in that interval do not belong to the domain of f, $\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{1}{\sqrt{x-1}}$ does not exist.

Chapter 2: Limit Theorems

Exercise 4.2.4

Prove that $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ does not exist but that $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0$.

Solution:

(I) Prove that $\lim_{x\to 0} \cos\left(\frac{1}{x}\right)$ does not exist.

Proof. Let (x_n) be the sequence $x_n = \frac{1}{2\pi n}$ and (y_n) be the sequence $y_n = \frac{1}{(2n-1)\pi}$. By algebraic limit rules, we know that $\lim_{n\to\infty}(x_n) = \lim_{n\to\infty}(y_n) = 0$. Now consider the limits of sequences $\cos\left(\frac{1}{(x_n)}\right)$ and $\cos\left(\frac{1}{(y_n)}\right)$ as n approaches infinity. We can see that:

$$\lim_{n\to\infty} \cos\left(\frac{1}{(x_n)}\right) = \lim_{n\to\infty} \cos(2\pi n) = 1 \qquad \lim_{n\to\infty} \cos\left(\frac{1}{y_n}\right) = \lim_{n\to\infty} \cos((2n-1)\pi) = -1$$

Therefore, since there exists two different sequences both converging to zero for which the function does not converge to the same result, by the sequential definition of functional limits, the limit does not exist. □

(II) Prove that $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0$.

Proof. Since $-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$, we can establish that $-x \leq x\cos\left(\frac{1}{x}\right) \leq x$. Now, since $\lim_{x\to 0} -x = \lim_{x\to 0} x = 0$, by the Squeeze Theorem, we know that $\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0$ □

Exercise 4.2.5

Let f, g be defined on $A \subseteq \mathbb{R}$ to $\mathbb{R}$, and let c be a cluster point of A. Suppose that f is bounded on a neighbourhood of c and that $\lim_{x\to c} g = 0$. Prove that $\lim_{x\to c} f \cdot g = 0$.

Solution:

Proof. Let δ_0 be the neighbourhood around c for which f is bounded. Thus, there exists some $M \in \mathbb{R}$ such that $M > 0$ for which $\forall x \in V_{\delta_0}(c)$, $|f(x)| < M$. Let $\epsilon > 0$ be arbitrary. Since $\lim_{x\to c} g = 0$, we know by the definition of functional limits that there exists some

δ_1 such that for all x satisfying $0 < |x-c| < \delta_1$, then g satisfies $|g(x) - 0| = |g(x)| < \frac{\epsilon}{M}$. Letting $\delta = \min \delta_0, \delta_1$, we therefore know that for all x satisfying $0 < |x - c| < \delta$:

$$\begin{aligned} |f(x)g(x) - 0| &= |f(x)g(x)| \\ &= |f(x)||g(x)| \\ &< M \cdot \frac{\epsilon}{M} \\ &= \epsilon \end{aligned}$$

Thus, since for arbitrary $\epsilon > 0$, there always exists some $\delta > 0$ for which x satisfying $0 < |x-c| < \delta$ implies $|f(x)g(x)-0| < \epsilon$, then by definition of functional limits, $\lim_{x\to c} f(x)g(x) = 0$. □

Exercise 4.2.8

Let $n \in \mathbb{N}$ be such that $n \geq 3$. Derive the inequality $-x^2 \leq x^n \leq x^2$ for $-1 < x < 1$. Then use the fact that $\lim_{x\to 0} x^2 = 0$ to show that $\lim_{x\to 0} x^n = 0$.

Solution:

(I) Derive the inequality $-x^2 \leq x^n \leq x^2$ for $-1 < x < 1$.

Since $n \geq 3$, we know that there exists some $k \in \mathbb{N}$ such that $n - k = 2$. Therefore, $x^n = x^2 x^k$. Since $-1 < x < 1$, we know that $-1 \leq x^k \leq 1$ and thus $-x^2 \leq x^2 x^k \leq x^2$. Therefore, since $x^n = x^2 x^k$, we can conclude that $-x^2 \leq x^n \leq x^2$.

(II) Show that $\lim_{x\to 0} x^n = 0$.

Using our prior inequality, we know that since $-x^2 \leq x^n \leq x^2$, thus, we know that $\lim_{x\to 0} -x^2 \leq x^n \leq \lim_{x\to 0} x^2$. Since we know that $\lim_{x\to 0} x^2 = 0$, by algebraic limit theorems, we can conclude that $\lim_{x\to 0} -x^2 = \lim_{x\to 0} x^2 = 0$, and thus, by squeeze theorem, we can conclude that $\lim_{x\to 0} x^n = 0$ for all $n \geq 3$.

Exercise 4.2.9

Let f, g be defined on A to $\mathbb{R}$ and let c be a cluster point of A.

(a) Show that if both $\lim_{x\to c} f$ and $\lim_{x\to c}(f + g)$ exists, then $\lim_{x\to c} g$ exists.

Solution:

Let L_1, $L_2 \in \mathbb{R}$ such that $\lim_{x\to c} f = L_1$ and $\lim_{x\to c}(f+g) = L_2$. By algebraic limit theorems, we know that: $\lim_{x\to c}(f + g) - \lim_{x\to c} f = L_2 - L_1$ and thus:

$$\begin{aligned} L_2 - L_1 &= \lim_{x\to c}(f+g) - \lim_{x\to c} f \\ &= \lim_{x\to c}(f+g-f) \\ &= \lim_{x\to c} g \end{aligned}$$

Thus, letting $L_3 = L_2 - L_1$, since there exists an $L_3 \in \mathbb{N}$ such that $\lim_{x\to c} g = L_3$, we know that $\lim_{x\to c} g$ exists.

(b) If $\lim_{x\to c} f$ and $\lim_{x\to c} f \cdot g$ exist, does it follow that $\lim_{x\to c} g$ exists?

Solution:

No. Take for example $f(x) = x^2$ and $g(x) = \frac{1}{x}$. Noting that $f \cdot g = x^2 \cdot \frac{1}{x} = x$, we conclude that $\lim_{x\to 0} f(x) = \lim_{x\to 0}(f(x)\cdot g(x)) = 0$. However, $\lim_{x\to c} g = \lim_{x\to c} \frac{1}{x}$ does not have a limit in the real numbers. Therefore, it does not follow that $\lim_{x\to c} g$ exists even if $\lim_{x\to c} f$ and $\lim_{x\to c}(f\cdot g)$ do.

Exercise 4.2.11

Determine whether the following limits exist in $\mathbb{R}$.

(a) $\lim_{x\to 0} \sin\left(\frac{1}{x^2}\right)$ $(x \neq 0)$.

Solution:

This limit does not exist. Consider the two sequences $(x_n) = \frac{1}{\sqrt{2\pi n + \frac{\pi}{2}}}$ and $(y_n) = \frac{1}{\sqrt{2\pi n + \frac{3\pi}{2}}}$. We can see that trivially see that as n approaches infinity, (x_n) and (y_n) approach 0. However, we can also observe that:

$$\begin{aligned} \lim_{n\to\infty} \sin\left(\frac{1}{(x_n)^2}\right) &= \lim_{n\to\infty} \sin\left(2\pi n + \frac{\pi}{2}\right) \\ &= 1 \end{aligned} \qquad \begin{aligned} \lim_{n\to\infty} \sin\left(\frac{1}{(y_n)^2}\right) &= \sin\left(2\pi n + \frac{3\pi}{2}\right) \\ &= -1 \end{aligned}$$

Thus, since there exists two sequences which both converge to zero but result in the function converging to two different limits, we know that $\lim_{n\to 0} \sin\left(\frac{1}{x^2}\right)$ does not exist.

(b) $\lim_{x\to 0} x \sin\left(\frac{1}{x^2}\right)$ $(x \neq 0)$.

Solution:

This limit does exist. Since $-1 \leq \sin\left(\frac{1}{x^2}\right) \leq 1$, we ascertain that $-x \leq \sin\left(\frac{1}{x^2}\right) \leq x$. Since $\lim_{x\to 0} -x = \lim_{x\to 0} x = 0$, by the squeeze theorem, we know that $\lim_{x\to 0} x \sin \frac{1}{x^2} = 0$.

(c) $\lim_{x\to 0} \operatorname{sign}\left(\sin\left(\frac{1}{x^2}\right)\right)$ $(x \neq 0)$.

Solution:

This limit does not exist. Once again, consider the two sequences $(x_n) = \frac{1}{\sqrt{2\pi n + \frac{\pi}{2}}}$ and $(y_n) = \frac{1}{\sqrt{2\pi n + \frac{3\pi}{2}}}$. Since both sequences converge to 0 as n goes to infinity, to show that $\lim_{x\to 0} \operatorname{sign}\left(\sin\left(\frac{1}{x^2}\right)\right)$ $(x \neq 0)$ does not exist, we can show that the function converges to two different values for each sequence. We know that $\sin\left(2\pi n + \frac{\pi}{2}\right) = 1$ for all $n \in \mathbb{N}$ and similarly, $\sin\left(2\pi n + \frac{3\pi}{2}\right) = -1$ for all $n \in \mathbb{N}$. Therefore:

$$\begin{aligned}
\lim_{n\to\infty} \operatorname{sign}\left(\sin\left(\frac{1}{(x_n)^2}\right)\right) &= \lim_{n\to\infty} \operatorname{sign}\left(\sin\left(2\pi n + \frac{\pi}{2}\right)\right) \\
&= \lim_{n\to\infty} \operatorname{sign}(1) \\
&= 1
\end{aligned}$$

$$\begin{aligned}
\lim_{n\to\infty} \operatorname{sign}\left(\sin\left(\frac{1}{(y_n)^2}\right)\right) &= \lim_{n\to\infty} \operatorname{sign}\left(\sin\left(2\pi n + \frac{3\pi}{2}\right)\right) \\
&= \lim_{n\to\infty} \operatorname{sign}(-1) \\
&= -1
\end{aligned}$$

Thus, since there exists two sequences which both converge to zero but result in the function converging to two different limits, we know that $\lim_{n\to 0} \operatorname{sign}\left(\sin\left(\frac{1}{x^2}\right)\right)$ does not exist.

(d) $\lim_{x\to 0} \sqrt{x}\sin\left(\frac{1}{x^2}\right)$.

Solution:

This limit does not exist, simply because there exists no open interval around $x = 0$ which is contained in the function's domain (since $\sqrt{x}$ is undefined for all $x < 0$).

This right-hand limit does however exist. Since $-1 \leq \sin\left(\frac{1}{x^2}\right) \leq 1$, we can ascertain that $-\sqrt{x} \leq \sin\left(\frac{1}{x^2}\right) \leq \sqrt{x}$. Since $\lim_{x\to 0^+} -\sqrt{x} = \lim_{x\to 0^+} \sqrt{x} = 0$, by the squeeze theorem, we know that $\lim_{x\to 0^+} \sqrt{x}\sin\frac{1}{x^2} = 0$.

Exercise 4.2.15

Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$, and let $c \in \mathbb{R}$ be a cluster point of A. In addition, suppose that $f(x) \geq 0$ for all $x \in A$, and let $\sqrt{f}$ be the function defined for $x \in A$ by $\left(\sqrt{f}\right)(x) := \sqrt{f(x)}$. If $\lim_{x\to c}$ exists, prove that $\lim_{x\to c} \sqrt{f} = \sqrt{\lim_{x\to c} f}$.

Solution:

Proof. Let $L = \lim_{x\to c} f$. By algebraic limit rules, we know that $\sqrt{L} = \lim_{x\to c} \sqrt{\lim_{x\to c} f}$. Since $\lim_{x\to c} f$ exists, then for arbitrary $\epsilon > 0$, there exists some $\delta > 0$ such that all $x \in A$ satisfying $0 < |x - c| < \delta$ also satisfy $|f(x) - f(c)| < \sqrt{L}\epsilon$. Therefore, for all x satisfying $0 < |x - c| < \delta$, we have that:

$$\begin{aligned}
|\sqrt{f} - \sqrt{L}| &= |\sqrt{f} - \sqrt{L}| \cdot \frac{\sqrt{f} + \sqrt{L}}{\sqrt{f} + \sqrt{L}} \\
&= \frac{|\left(\sqrt{f}\right)^2 - L|}{\sqrt{f} + \sqrt{L}} \\
&= \frac{|f - L|}{\sqrt{f} + \sqrt{L}} \\
&\leq \frac{|f - L|}{\sqrt{L}} \\
&< \frac{\sqrt{L}\epsilon}{\sqrt{L}} \\
&= \epsilon
\end{aligned}$$

Therefore, since for arbitrary $\epsilon > 0$, there exists some $\delta > 0$ for which all x satisfying $0 < |x - c| < \delta$ also satisfies $|\sqrt{f} - \sqrt{L}| < \epsilon$, by definition of functional limits, $\lim_{x\to c} \sqrt{f} = \sqrt{L}$. Since $\sqrt{L} = \sqrt{\lim_{x\to c} f}$, we thus have that:

$$\lim_{x\to c} \sqrt{f} = \sqrt{\lim_{x\to c} f}$$

□

Chapter 3: Some Extensions of the Limit Concept

Exercise 4.3.4

Let $c \in \mathbb{R}$ and let f be defined for $x \in (c, \infty)$ and $f(x) > 0$ for all $x \in (c, \infty)$. Show that $\lim_{x\to c} f = \infty$ if and only if $\lim_{x\to c} \frac{1}{f} = 0$.

Solution:

(I) Forward direction.

If $\lim_{x\to c} f = \infty$, then we know that for all $M > 0$, there exists some $\delta > 0$ such that all x satisfying $0 < |x - c| < \delta$ also satisfies $f(x) > M$. Let $\epsilon = \frac{1}{M}$, we can therefore see that, $\forall x \in V_\delta(c)$:

$$\begin{aligned} &f(x) > M \implies \\ &f(x) > \frac{1}{\epsilon} \end{aligned}$$

Since $f(x) > 0$ for all $x \in (c, \infty)$, we know that $f(x) = |f(x)|$. Therefore, we can establish that:

$$\begin{aligned} f(x) > \frac{1}{\epsilon} &\implies \\ |f(x)| > \frac{1}{\epsilon} &\implies \\ \frac{1}{|f(x)|} &< \epsilon \\ \left|\frac{1}{f(x)}\right| &< \epsilon \\ \left|\frac{1}{f(x)} - 0\right| &< \epsilon \end{aligned}$$

Therefore, since for all $\epsilon > 0$, there exists some $\delta > 0$ such that all x satisfying $0 < |x-c| < \delta$ also satisfies $|\frac{1}{f(x)} - 0| < \epsilon$, by definition of functional limits, we know $\lim_{x\to c} \frac{1}{f} = 0$.

(II) Reverse Direction.

Let $\epsilon > 0$

Exercise 4.3.8

Let f be defined on $(0, \infty)$ to $\mathbb{R}$. Prove that $\lim_{x\to\infty} = L$ if and only if $\lim_{x\to 0^+} f\left(\frac{1}{x}\right) = L$.

Solution:

Exercise 4.3.12

Find functions f and g defined on $(0, \infty)$ such that $\lim_{x\to\infty} f = \infty$ and $\lim_{x\to\infty} = \infty$ and $\lim_{x\to\infty}(f - g) = 0$. Can you find such functions, with $g(x) > 0$ for all $x \in (0, \infty)$ such that $\lim_{x\to\infty} \frac{f}{g} = 0$?

Solution:

Exercise 4.3.13

Let f and g be defined on (a, ∞) and suppose that $\lim_{x\to\infty} f = L$ and $\lim_{x\to\infty} g = \infty$. Prove that $\lim_{x\to\infty} f \circ g = L$.